

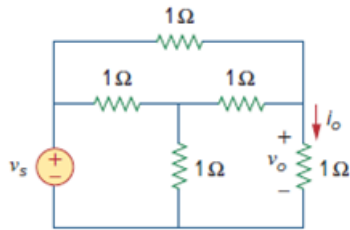
Problem

(a) In the circuit of Fig. 4.71, calculate v_o and i_o when $v_s = 1\text{ V}$.

(b) Find v_o and i_o when $v_s = 10\text{ V}$.

(c) What are v_o and i_o when each of the $1\text{-}\Omega$ resistors is replaced by a $10\text{-}\Omega$ resistor and $v_s = 10\text{ V}$?

Figure. 4.71:



Step-by-step solution

Step 1 of 9

(a)

Refer to Figure 4.71 in the textbook.

Redraw the circuit as shown in Figure 1.

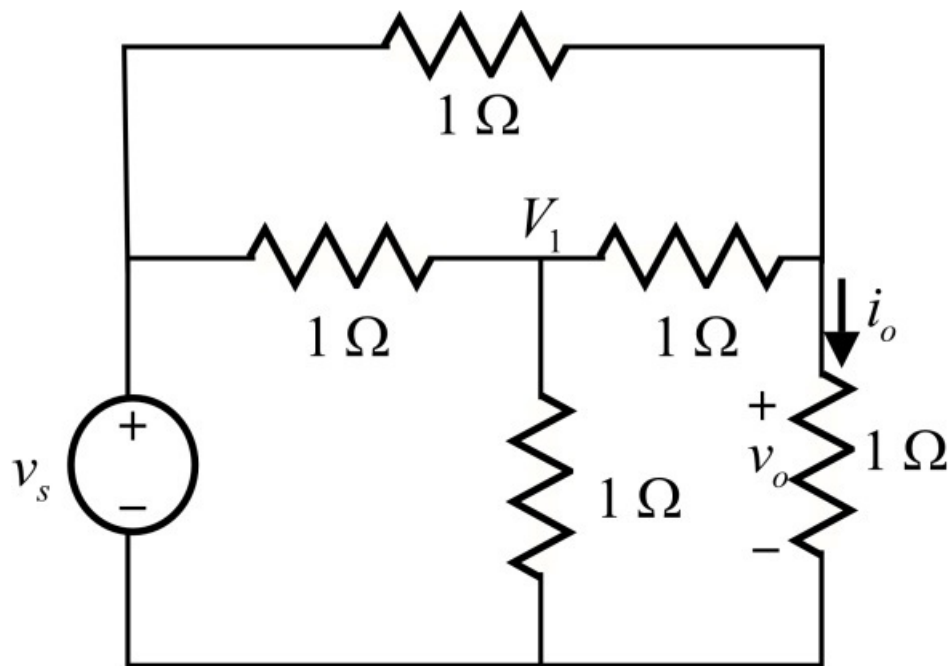


Figure 1

Step 2 of 9

Apply Kirchhoff's current law at node v_o .

$$\frac{v_o - V_1}{1\ \Omega} + \frac{v_o - v_s}{1\ \Omega} = -\frac{v_o}{1\ \Omega}$$

$$v_o - V_1 + v_o - v_s = -v_o$$

$$3v_o - v_s = V_1$$

$$V_1 = 3v_o - v_s \dots\dots (1)$$

Apply Kirchhoff's current law at node V_1 .

$$\frac{V_1 - v_s}{1\ \Omega} + \frac{V_1}{1\ \Omega} + \frac{V_1 - v_o}{1\ \Omega} = 0$$

$$V_1 - v_s + V_1 + V_1 - v_o = 0$$

$$v_o = 3V_1 - v_s \dots\dots (2)$$

Step 3 of 9

Substitute equation 1 in equation 2.

$$v_o = 3V_1 - v_s$$

$$v_o = 3[3v_o - v_s] - v_s$$

$$v_o = 9v_o - 3v_s - v_s$$

$$8v_o = 4v_s$$

$$v_o = \frac{4}{8}v_s \dots\dots (3)$$

Step 4 of 9

Consider $v_s = 1\text{ V}$.

Determine the output voltage v_o .

Substitute 1 V for v_s in equation 3.

$$v_o = \frac{4}{8}v_s$$

$$= \frac{4}{8}(1\text{ V})$$

$$= 0.5\text{ V}$$

Determine the current i_o .

Consider the equation for i_o .

$$i_o = \frac{v_o}{1\ \Omega} \dots\dots (4)$$

Substitute 0.5 V for v_o in equation 4.

$$i_o = \frac{v_o}{1\ \Omega}$$

$$= \frac{0.5\text{ V}}{1\ \Omega}$$

$$= 0.5\text{ A}$$

Therefore, the output voltage v_o is 0.5 V and the current i_o is 0.5 A .

Step 5 of 9

(b)

Consider $v_s = 10 \text{ V}$.

Determine the output voltage v_o .

Substitute 10 V for v_s in equation 3.

$$\begin{aligned} v_o &= \frac{4}{8} v_s \\ &= \frac{4}{8} (10 \text{ V}) \\ &= 5 \text{ V} \end{aligned}$$

Determine the current i_o .

Consider the equation for i_o .

$$i_o = \frac{v_o}{1 \Omega}$$

Step 6 of 9

Substitute 5 V for v_o in equation 4.

$$\begin{aligned} i_o &= \frac{v_o}{1 \Omega} \\ &= \frac{5 \text{ V}}{1 \Omega} \\ &= 5 \text{ A} \end{aligned}$$

Therefore, the output voltage v_o is 5 V and the current i_o is 5 A .

Step 7 of 9

(c)

Replace 1Ω resistors by 10Ω resistors in Figure 1.

Apply Kirchhoff's current law at node v_o .

$$\frac{v_o - V_1}{10 \Omega} + \frac{v_o - v_s}{10 \Omega} = -\frac{v_o}{10 \Omega}$$

$$\frac{v_o - V_1 + v_o - v_s}{10} = \frac{-v_o}{10}$$

$$3v_o - v_s = V_1$$

$$V_1 = 3v_o - v_s \dots\dots (5)$$

Apply Kirchhoff's current law at node V_1 .

$$\frac{V_1 - v_s}{10 \Omega} + \frac{V_1}{10 \Omega} + \frac{V_1 - v_o}{10 \Omega} = 0$$

$$\frac{V_1 - v_s + V_1 + V_1 - v_o}{10} = \frac{v_o}{10}$$

$$v_o = 3V_1 - v_s \dots\dots (6)$$

Step 8 of 9

Substitute equation 5 in equation 6.

$$v_o = 3V_1 - v_s$$

$$v_o = 3[3v_o - v_s] - v_s$$

$$v_o = 9v_o - 3v_s - v_s$$

$$8v_o = 4v_s$$

$$v_o = \frac{4}{8}v_s \dots\dots (7)$$

Step 9 of 9

Determine the output voltage v_o .

Substitute 10 V for v_s in equation 7.

$$\begin{aligned} v_o &= \frac{4}{8}v_s \\ &= \frac{4}{8}(10 \text{ V}) \\ &= 5 \text{ V} \end{aligned}$$

Determine the current i_o .

Consider the equation for i_o .

$$i_o = \frac{v_o}{10 \Omega} \dots\dots (8)$$

Substitute 5 V for v_o in equation 8.

$$\begin{aligned} i_o &= \frac{v_o}{10 \Omega} \\ &= \frac{5 \text{ V}}{10 \Omega} \\ &= 0.5 \text{ A} \end{aligned}$$

Therefore, the output voltage v_o is 5 V and the current i_o is 0.5 A.